

Adnan Ahmed Khan

AI Engineer | LLM Agents, Agentic Workflows, Python & AWS

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PROFESSIONAL SUMMARY

AI Engineer and backend/platform specialist with 6+ years building reliable Python, TypeScript, AWS and distributed systems. Recent work focuses on LLM agents, graph-based workflows, prompt engineering, tool calling, structured outputs, model routing and production-oriented AI automation. Combines hands-on LangGraph, LangChain, CrewAI, Pydantic and OpenClaw engineering with a track record scaling platforms to 500-800 RPS and 50k-100k DAU, reducing failed operations 80%, AWS spend 25% and MTTR from about 45 minutes to under 10.

TECHNICAL SKILLS

LLM & Agent Engineering: LLM applications, AI agents, multi-agent workflows, prompt engineering, tool calling, MCP/tool integrations, structured JSON outputs, workflow orchestration, model routing, evaluation and testing

AI Frameworks: LangGraph, LangChain Core/Workflows, CrewAI, OpenClaw, GitHub Copilot API, Pydantic, pydantic-settings, ChromaDB

Languages & APIs: Python 3.12, TypeScript, JavaScript, FastAPI, Node.js, Nest.js, Express, REST APIs, async HTTP/httpx

Cloud & Platform: AWS, EKS, ECS, Lambda, Docker, Kubernetes, Terraform, Helm, GitHub Actions, ArgoCD

Data & Reliability: PostgreSQL, Redis, Kafka, Elasticsearch, MongoDB, DynamoDB, idempotency, retries, distributed state, encryption, pytest, Jest/node:test, Prometheus, Grafana, Jaeger

SELECTED AI & LLM PROJECTS

openclaw-workflow - Declarative Multi-Agent Workflow Orchestrator | [GitHub](#)

- Architected a YAML/JSON-driven orchestration runtime for LLM agent pipelines with dependency graphs, parallel execution, retries, timeouts, loop expansion, optional/always-run steps, cancellation and partial resume.
- Implemented per-step model overrides, runtime/legacy/CLI session adapters, durable filesystem/native Redis/MCP Redis state, status notifications and output gates that verify artifacts before downstream steps execute.
- Added schema and contract validation with TypeBox, AJV and CEL plus automated coverage for adapters, semantic state contracts, output gating, state backends and workflow execution.

_tasks - Python Agentic Automation Platform | [GitHub](#)

- Built a Python 3.12 agent automation platform using LangGraph, LangChain Core, CrewAI and Pydantic with a supervisor/worker/reducer architecture, GitHub Actions execution and browser-driven task graphs.
- Developed an asynchronous LLM client supporting chat completions and tool-calling/reasoning endpoints, dynamic model discovery, quota-aware cost routing, transcript logging and structured outputs.
- Created agent graphs for job application, browser navigation and form filling; prompt pipelines for relevance scoring and ATS resume tuning; and durable shared-state coordination with encrypted secrets, leases and presence records.

PROFESSIONAL EXPERIENCE

HackOnTech | Senior Software Engineer, Backend & Platform | Remote / Islamabad | 2023 - Present

- Own core backend and platform engineering for B2B/G2G NGO systems serving 50k-100k daily users and 500-800 RPS peak traffic.
- Led a 0-to-1 platform overhaul into six high-throughput microservices, enabling zero-downtime releases while protecting payments, queues and customer-facing availability.
- Engineered resilient APIs and distributed workflows with idempotency, retries, rollbacks and recovery, cutting failed cross-service operations 80% and payment failures from about 5% to 2%.
- Reduced high-volume PostgreSQL/Redis query latency from 30+ seconds to under 1 second through query redesign, indexing and caching.
- Reduced AWS spend 25% across EKS/autoscaling workloads and cut MTTR from about 45 minutes to under 10 with tracing, dashboards and service-level observability.

Environment: Python, FastAPI, TypeScript, Node.js, Nest.js, AWS EKS/Lambda, PostgreSQL, Redis, Temporal, Step Functions, Terraform, Helm, ArgoCD, Prometheus, Grafana, Jaeger

Creative IT Park | Backend Developer | 2021 - 2023

- Delivered and maintained Python/FastAPI and Node.js backend platforms for government, university and healthcare clients, owning production REST APIs and integrations.
- Hardened Kafka notification pipelines to preserve delivery during downstream failures and optimized data access/caching to reduce public-portal search latency.
- Secured healthcare ETL workflows with encrypted data flows and fine-grained AWS IAM controls, building experience in regulated and sensitive-data environments.

Environment: Python, FastAPI, Node.js, Kafka, Elasticsearch, Redis, AWS ECS/RDS/DynamoDB/Glue/IAM, PostgreSQL, MongoDB, Docker

GeekGenix | Backend Developer | 2020 - 2021

- Built Express/MongoDB APIs, Stripe checkout and webhook reconciliation workflows, and Passport.js authentication.

EDUCATION

BS Computer Science - PMAS Arid Agriculture University, Rawalpindi